Trim DAC Design with Minimum DNL for Self-Trim with Self-Test Schemes

Hans Martin von Staudt
Dialog Semiconductor
Swindon, United Kingdom
HansMartin.vonStaudt@diasemi.com

Abstract—**High precision mixed-signal design strongly depends on trimming techniques to overcome manufacturing induced variability, of which the digital trimming is the most popular approach. The central element of a digitally driven trim circuit is the trim DAC. Whilst it needs to be simple and cheap its electrical performance in the digital to analogue conversion process will affect the trim result. In this paper we analyse the trimmed distribution as a function of the DNL performance of the trim DAC. Of particular concern is the shaping of the distribution by comparator based self-calibration schemes. A minimum DNL trim DAC design is proposed that overcomes unwanted distribution tails. Eliminating distribution tails is a design goal for any type of trimming but it is crucially important if an on-chip self-test scheme shall be added to the self-trimming. A simulation model for the trimmed distribution is compared to actual silicon results of standard and minimum-DNL trim DACs.**

Keywords—self-trim by reference, self-calibration, self-trimming, self-test, self-tuning, trim DAC

I. INTRODUCTION

The fundamental trend in microelectronics to use smaller and smaller devices increases inevitably variability. In principle this effect is well known already in analogue circuit design. As the absolute electrical variation of semiconductor devices is huge (~30%) most precision circuits are based on matching performance i.e. the ability to process two devices that match each other even though both undergo the same absolute variation. It is determined by the geometric variability of the process which manifests itself at the fringes of each shape. By increasing the size of corresponding shapes the matching performance increases as the fringes compared to the total area become negligible.

This constitutes however a conflict with the commercial requirement for small chip size. A long standing practice therefore is trimming. The shapes are designed significantly smaller than ideal but are equipped with a small adjustable element as depicted in Figure 1.

The performance of the circuit is measured during the mandatory manufacturing step of testing and the adjustable element is trimmed (by switching on or off small sub-elements) until the desired value is reached. The same type of digitally driven trimming is used for reference circuits [1].



Fig. 1. Large matching structures vs. small structures with trim elements

In theory the trend to smaller geometries should help the geometric variability. However, electrical variability (threshold voltage Vth, saturation current IDsat) becomes dominating. The requirement for trimming becomes ubiquitous and shifts the focus from static external trimming to a dynamic on-chip process. Self-calibration for matching [2], against an external reference [3, 4], or against an indirect specification [5, 6] has been shown to be an attractive approach to tackle variability[1].

On-chip self-calibration raises the requirement for self-test. In [7] a window test was proposed that checks against a simple fault model for the comparator based self-trim. The test scheme checks the trimmed distribution not to be much wider than the rectangle 1 LSB wide. However, as shown in [7] the imperfections of the comparator and in particular of the trim DAC potentially result in long distribution tails, which can easily fail the window test. Hence it is important to have a mathematical way to describe it.

II. TRIMMED DISTRIBUTION

A. Ideal Trimming Process

The ideal trimming process divides the natural distribution of the value, characterised by its standard deviation σ_V, in a series of usually 2^n slices, each 1 LSB wide. As depicted in Figure 2 the trimming process piles the slices on top of each other.

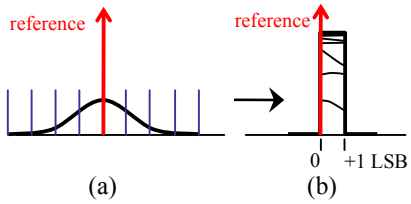


Fig. 2. Natural distribution cut in slices, forming a trimmed rectangle

For the remainder of this paper we assume that the trim range $R = 2^n \cdot LSB$ is designed wide enough to cover the

[1] The terms calibration and trimming are used interchangeably. There is no particular convention in this paper to distinguish between the two terms.

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natural distribution of the value to be trimmed, usually $\pm 5\sigma_V$, so the tails beyond R can be neglected. As a further convention we further assume that the centre of the untrimmed distribution $\mu = 0$, as well as the target value of the (self-) trim operation. The trim codes may be designed in a 2's complement form with $i \in [-2^{n-1}, 2^{n-1}-1]$. Each slice can then be described as follows.

(1)
$$slice(x,i) = \begin{cases} \frac{1}{\sqrt{2\pi\sigma_V^2}} \cdot e^{-\frac{\left(x - iR/2^n\right)^2}{2\sigma_V^2}} & \text{for } i\frac{R}{2^n} \le x < (i+1)\frac{R}{2^n} \\ 0 & \text{otherwise} \end{cases}$$

It can be shown that the sum of all slices form the ideal rectangular trimmed distribution as depicted in Figure 2b. The second line expresses the rectangular distribution with the step function (Heaviside function [8]) *H(x)*.

(2)
$$V(x) = \sum slice(x,i) = \begin{cases} \frac{1}{LSB} & \text{for : } 0 \ge x > LSB \\ 0 & \text{otherwise} \end{cases}$$

(3)
$$V(x) = \frac{H(x) - H(LSB - x)}{LSB}$$

We assume here the comparator-based self-trim, which stops once the value to be trimmed is beyond the reference. This affects the rectangle such that it is not centred on the reference but single-sided attached to it (see Figure 2.).

B. *Trimming Process With Noise*

The measurement process required for trim decisions is a second independent random process, this time in the time domain. In the example of the comparator-based self-trim it is the uncertainty if the difference between the reference and the value under trim is close to zero. Under this condition any kind of noise will determine the outcome of the comparison. Other terms for the same effect are repeatability and random measurement inaccuracy. This random process can be described with the probability density function (PDF) $\mathcal{N}(\mu=0, \sigma_N)$. As depicted in Figure 3 each transition is affected by random noise.

It shapes the form of each of the slices. The mathematical operation expressing this is a convolution [9], indicated by the asterisk. As the convolution is a linear operation of its application to all slices is equivalent to applying it to the ideal sum of all slices, which is the rectangle

(4)
$$\begin{aligned} V(x) &= \sum \left(N(0,\sigma_N) * slice(x,i) \right) \\ &= N(0,\sigma_N) * \sum slice(x,i) \\ &= N(0,\sigma_N) * \frac{H(x) - H(LSB - x)}{LSB} \end{aligned}$$

The result of convoluting a step function with a PDF is equal to the cumulative distribution function (CDF). For a normal distribution the CDF is expressed with the error function erf(x) [10].

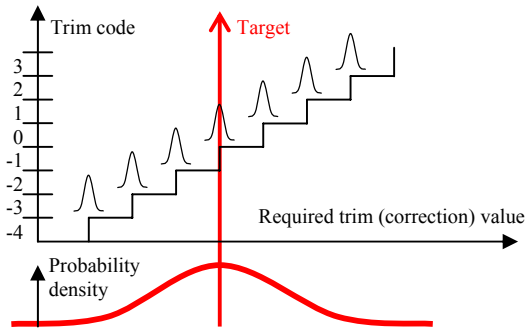


Fig. 3. Trim DAC transfer function, modified by noise

(5)
$$V(x) = \frac{erf\left(\frac{x}{\sqrt{2\sigma_N^2}}\right) - erf\left(\frac{x - LSB}{\sqrt{2\sigma_N^2}}\right)}{2LSB}$$

C. *Trimming Process With Differential Non-Linearity*

The effect of manufacturing variations on the elements of a DAC have been widely examined in [11, 12] with regards to DNL and INL performance and yield. A trim DAC, however, has only one purpose: to find a (i.e. any) correction of the untrimmed value. Hence INL is of no particular interest as it is overshadowed by the width of the untrimmed distribution.

In contrast DNL refers to the width of the trimmed distribution as it describes the ability of a DAC to target a given output value. For trim DACs this effect is expected to become visible in the trimmed distribution.

The definition of the slices in (1) needs to be changed in order to take the variation of their widths into account. As discussed above the INL contribution in (6) can be neglected as it will be eliminated as part of the trimming process.

(6)
$$slice(x,i) = \begin{cases} 0 & \text{for } x \le i\frac{R}{2^n} + INL(i) \\ 0 & \text{for } x > (i+1)\frac{R}{2^n} + INL(i) + DNL(i+1) \\ \frac{1}{\sqrt{2\pi\sigma_V^2}} \cdot e^{-\frac{(x-i)^2}{2\sigma_V^2}} & \text{otherwise} \end{cases}$$

The width of each step is determined by the corresponding DNL(i+1), which again can be described with a normally distributed probability density function, similar to what was depicted in Figure 3. A convolution between the two PDFs would be required.

There are two major differences to the convolution with the comparator noise. Firstly the self-trim algorithm eliminates the left hand tail of the distribution, because it carries on stepping to the next slice until the trimmed value is beyond the reference. The tail which is cut off on the left hand side will be

shifted with the next trim step by 1 LSB and will then be subjected to a convolution operation before its contribution would be added. For the sake of simplicity we neglect this small contribution to the target rectangle and rather focus on the tail which is produced by the last self-trimming step.

As a second simplification we set the PDF of the untrimmed distribution to be constant for the width of the slice. Then each of the slices can be described as the convolution of a step function with the normal distribution of the respective DNL(i).

$$slice(x,i) = \begin{cases} 0 & \text{for } x < 0 \\ \underbrace{\frac{1}{\sqrt{2\pi\sigma_V^2}} \cdot e^{-\frac{\left(iR/2^n\right)^2}{2\sigma_V^2}}}_{\text{Probability of transition}} \cdot \underbrace{\frac{1}{2}\left(1 - erf\left(\frac{x - LSB}{\sqrt{2\sigma_i^2}}\right)\right)}_{\text{CDF of DNL(i)}} & \text{for } x \geq 0 \end{cases} \quad (7)$$

The total trim distribution of the values is again the sum of all slices. $V(x) = \sum slice(x,i)$.

The second major difference to the trimming process with noise is that each slice is characterised by a different σ_i. Hence the terms can't be simplified but remain a sum of terms. σ_i depends on the number of simultaneously switching trim elements. It is well known that especially for binary weighted DACs the major transition gives the highest DNL contribution. Table I compares the DNL values of the transitions of a 3-bit DAC (in brackets: 4-bit DAC) for binary weighted and thermometer encoded DACs. It is assumed that both DACs are made of unit elements, each with a standard deviation of σ_U.

TABLE I. DNL OF 3(4)-BIT TRIM DAC TRANSITIONS

Transition:	**1 (15)**	**2 (14)**	**3 (13)**	**4 (12)**	**5 (11)**	**6 (10)**	**7 (9)**	**(8)**
Elements switching	1	3	1	7	1	3	1	15
DNL(i) Binary weights	σ_U	$\sigma_U \cdot\sqrt{3}$	σ_U	$\sigma_U \cdot\sqrt{7}$	σ_U	$\sigma_U \cdot\sqrt{3}$	σ_U	$\sigma_U \cdot\sqrt{15}$
DNL(i) Themometer	σ_U	σ_U	σ_U	σ_U	σ_U	σ_U	σ_U	σ_U

It is no surprise that the binary weighted DAC shows strong DNL peaks at the major transitions. However, a binary DAC usually delivers the easiest and most area efficient implementation.

Depicted in Figure 4 are the PDFs of the slices associated to the major transition (4), its two neighbours (3, 5), and the trimmed distribution *V(x)* as the sum of all slices. At first sight *V(x)* looks like a normal CDF but the steepness of the slope at +1 LSB is followed by an unusually long tail. The explanation for this long tail is down to the rather high standard deviation of the major transition $\sigma_U \cdot\sqrt{(2n-1)}$. In fact, for a properly centred untrimmed distribution (see assumptions above) hitting the major transition is the most likely outcome of the self-trim process. The steepness of the slope is down to all the minor transitions with DNL(i) = σ_U.

III. DESIGN OF A MINIMUM-DNL TRIM DAC

As explained in II the left hand side tail is eliminated by the trim algorithm. Only the tail on the right hand side of the distribution is of concern. It is caused by the positive DNL of any transition i.e. the probability that the required trim step for reaching beyond the reference is too wide, hence ending up in the tail.

This observation is the basis for the suggestion to build a skewed trim DAC with a strong negative DNL. Under normal circumstances an intentionally "bad" design of a DAC would be avoided, but linearity is not a design goal for a trim DAC; actually it is only the first of the two goals below. The second is derived from our observation.

1. Ability to find one (or more) correction values inside the target distribution.
2. Minimise positive DNL to ensure a tail-free trim distribution

A. *Continuous Under-Dimensioning of Trim Elements*

One way to achieve both goals is to under-dimension every more significant trim element. Instead of a factor of 2.0 between trim elements a small fraction e.g. 5% is subtracted each time. Assume 5% is also the standard deviation σ_U of the unit element and $\sigma_n = \sigma_U \cdot\sqrt{(2n-1)}$ is the standard deviation of the major transitions.

TABLE II. KEY FIGURES OF 5% UNDER-DIMENSIONED TRIM DAC

	MSB					**LSB**
Nominal(n)	32	16	8	4	2	1
Skewed	24.76	13.03	6.86	3.61	1.90	1.00
DNL/LSB	-2.64	-1.34	-0.65	-0.29	-0.10	
DNL/σ_U	-52.8	-26.7	-13.0	-5.8	-2.0	
DNL/σ_n	-6.7	-4.8	-3.4	-2.2	-1.2	
Range Loss	-19%	-15%	-11%	-7%	-3%	

Table II illustrates the increasingly strong negative DNL of the transitions which suppresses the associated positive DNL tails for trim DACs up to 6 bits. In fact, as the major transitions are defused, the unsuppressed positive DNL of the minor transitions become the strongest contributor.

An obvious drawback of the skewed trim DAC design is the reduction in trim range. According to Table II it is moderate but noticeable. If that endangers the proper covering of the untrimmed distribution the trim range must be widened, either by increasing the LSB (for the price of a broader target distribution, here 19%), or by adding another trim bit i.e.

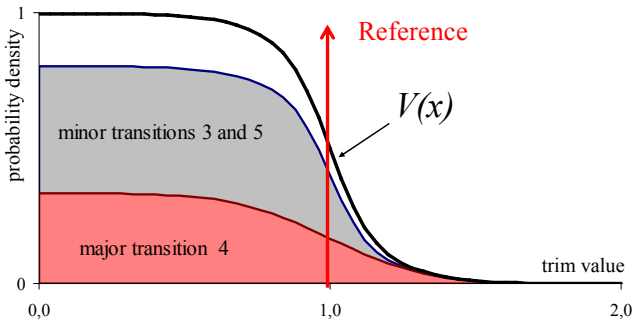


Fig. 4. DNL tail of major transistion dominates tail of trimmed distribution

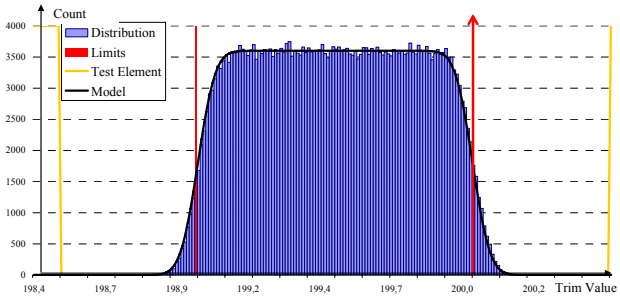


Fig. 5a. Simulation of comparator noise, $\sigma_N = 0.05$ LSB

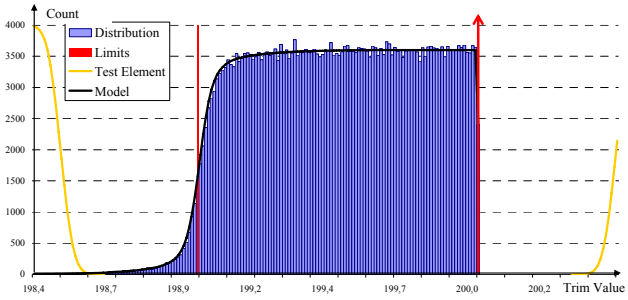


Fig. 5b. Simulation of DNL variation $\sigma_u = 0.05$ LSB, $\sigma_u = 4$ LSB

doubling the trim range. Of course this increases the required area but it does not come with the usual penalty of poorer positive DNL performance.

B. Discrete Under-Dimensioning of Trim Elements

In practical design the use of unit elements is prevalent. Hence a continuous under-dimensioning is not always easy to achieve (e.g. more difficult for current sources than for resistors). A variation of the continuous scheme according to Table II would be to focus only on the major transitions and to "discount" 1 element for each more significant bit. If the discounting would start already with the second trim element the trim DAC would degenerate and become thermometer encoded as all trim elements are equal.

TABLE III. KEY FIGURES OF "-1" UNDER-DIMENSIONED TRIM DAC

	MSB					**LSB**
Nominal(n)	32	16	8	4	2	1
Skewed	25	13	7	4	2	1
DNL/LSB	-3	-2	-1	0	0	
DNL/σ_U	-60	-40	-20	0	0	
DNL/σ_n	-8.3	-7.7	-5.3	0.0	0.0	
Range Loss	-17%	-13%	-7%	0%	0%	

In the example of Table III (again with σ_U = 5%) the suppression of the positive DNL is even stronger than for the three major transitions. The range loss is comparable but the price to pay for the discrete under-dimensioning are the untouched minor transitions which will push the trimmed distribution tail uncompensated.

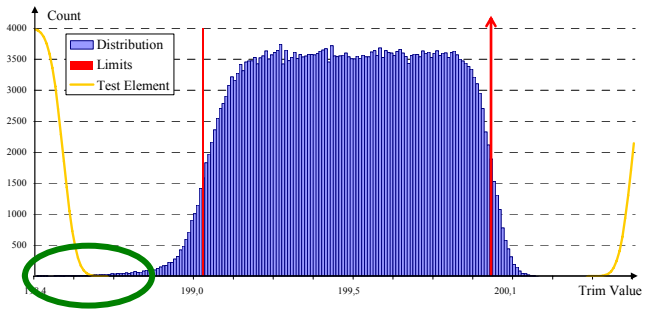


Fig. 6a. Simulation of comparator noise and DNL

IV. SIMULATION

A Monte-Carlo simulation of the statistical effects of noise and DNL distribution together with the original untrimmed distribution is the safest way to verify the design of any trim DAC in the presence of the aforementioned random processes. It can also be used to verify the analytical models derived in II.

As a particular concern the simulator evaluates the simulated tails and calculates possible yield loss against the self-test elements as introduced in [7]. It should be noted that in contrast to the figures in the previous sections the simulated system was designed to produce a left sided distribution (searching from above).

Figure 5 contains histograms of 2 simulations with 6-bit trim DACs against a natural untrimmed distribution with a standard deviation of 4 LSB and a target value of 200 LSB: The trimmed distribution is shaped by the comparator noise (a), and by the DNL distribution (b). The black curve is the analytical model according to II.B and II.C. It shows a very good fit and verifies the two methods against each other.

Figure 6 shows two further histograms for which no analytical model was derived. Figure 6a shows the trimmed distribution with both effects. Intuitively it looks like the convolution of the two graphs of Figure 5a and 5b. The simulated yield loss at the overlap of the 1.5 LSB test element and the tail represents already a fairly high yield loss of 300ppm. This might sound low but for practical implementations with a high number of such elements (see section V.) this value is critical. It is highlighted with the green ellipse in Figure 6a.

Figure 6b shows the design with the same random variables but with a "discounted" trim DAC for the 3 most significant

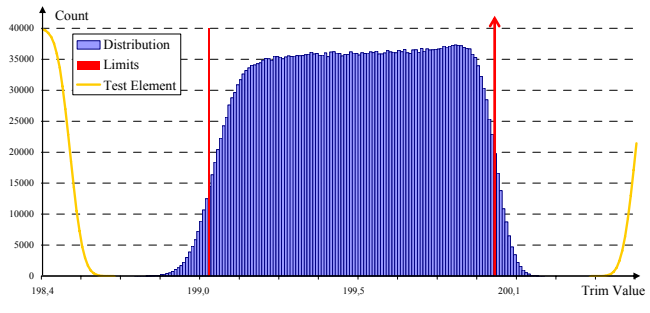


Fig. 6b. Simulation of "discounted" trim DAC with comparator noise and DNL

bits, as suggested in Table III. The suppression of the DNL tail is sufficient to push the yield loss to 0, even with a high number of identical self-calibrated and self-tested systems per chip.

It should be noted that the trim code distribution is no longer a direct representation of the untrimmed distribution. The negative DNL "discounts" create gaps in DAC's transfer function which in turn show up in the distribution of the trim codes (see Figure 7).

For all simulations no difference was ever encountered between binary and linear search. Linear search is referenced a couple of times throughout this paper as it is easier to grasp the effects of the linear search, for the critical moment the algorithm makes a decision to stop or to carry on searching the trip point.

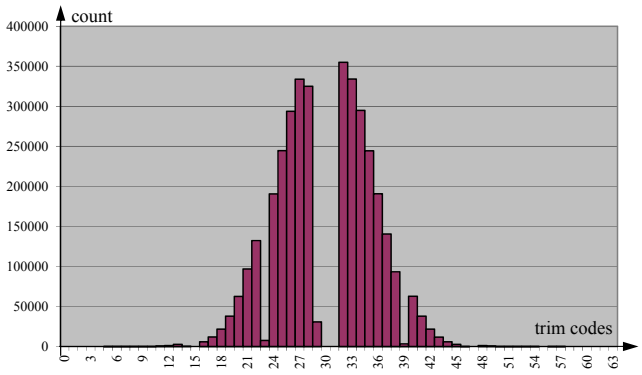


Fig. 7. Discontinuous trim code distribution of "discounted" trim DAC

V. Silicon Results

The techniques described above were applied to a passive matrix OLED driver chip with a display resolution of 240 x 320 RGB. To the display panel it features 720 precision current sources which need to be matched against each other within a ±0.25% range. For this design the self-trim and self-test algorithm was implemented but the first silicon showed huge tails which eventually failed most chips in the self-test phase of the self-trim.

The plots in Figure 8 compare the self-trim performance of the first and the second silicon iteration. On the second silicon only the MSB was discounted. This proved sufficient to suppress the DNL tail on the left hand side of the trimmed distribution. The remaining tails in Figure 8b are balanced which proves they are caused by a random noise process: either by the comparator noise or by the measurement noise.

Measurement noise was considerable when acquiring the values for Figure 8. It can be measured as repeatability yet despite heavy filtering it was not possible to eliminate it from the data. In fact, it constitutes a further convolution washing out the trimmed distribution (see Figure 6) further.

A contributor to the unsatisfying results on the first silicon iteration was the trim range being significantly wider than the untrimmed distribution. Consequently the positive DNL of the overly big major transition pushed the tail of the trimmed distribution. Nevertheless the overly large trim range design was exploited on the second silicon revision as only the MSB was discounted i.e. only the major transition was defused. Centring the trim range and the untrimmed distribution also eliminated the 2nd major transition as the untrimmed distribution was narrow enough not to touch these transitions. Only the 3rd major transition was touched at the fringes of the untrimmed distribution. The remaining minor transitions were anyhow unavailable for the "discounting" technique without dramatically limiting the trim range. Furthermore their contributions disappear behind the comparator noise which dominates Figure 8b.

VI. Conclusion

The design of an intentionally "bad" DAC, strongly skewed towards a negative differential non-linearity, makes it the best possible trim DAC. It proves to be a very simple yet effective way to suppress the distribution tail which results from the positive DNL of a trim DAC. The design scheme allows the unchanged use of trim algorithms for the comparator based self-trim and ensures the success of a window self-test scheme.

The disadvantage of a reduced trim range can be overcome by adding a trim bit without the risk normally associated with introducing a further more significant transition of a binary trim DAC.

The remaining risk of the positive DNL tail can be calculated with the developed formulas, if the untrimmed distribution is known. Hence a design-yield trade-off can be made to find an acceptable yield with the smallest overhead in terms of trim DAC modifications.

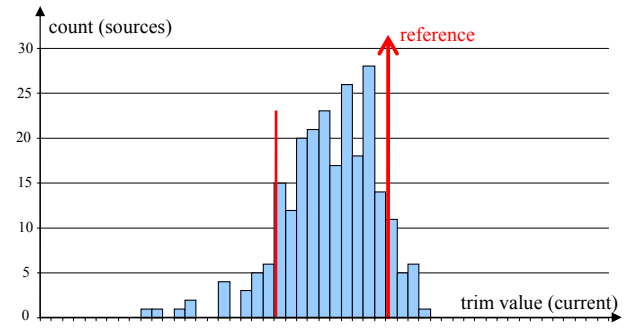


Fig. 8a. Trim distribution result of original trim DAC

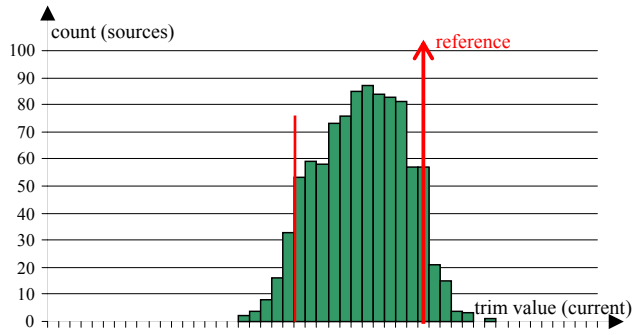


Fig. 8b. Trim distribution result of "discounted" trim DAC

The discounting of unit elements instead of continuously under-designed trim elements will be the practical implementation for most trim DACs.

Still a Monte-Carlo simulation is required to model the simultaneous effects of comparator noise, DNL tails and discounted trim elements. All effects analysed separately show a perfect match between the analytical solution and the simulation. An acceptable match to silicon performance can be achieved after extracting the standard distribution parameters.

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